UNIT QUESTIONS

SKILLS CRITICAL THINKING 3

1 The table shows the percentage of gases in inhaled and exhaled air.

Gas	Inhaled air / %	Exhaled air / %
nitrogen	78	79
oxygen		
carbon dioxide		
other gases (mainly argon)	1	1

a Copy the table and fill in the gaps by choosing from the following numbers:
21 4 0.04 16 **(2)**

SKILLS REASONING 6

b Explain why the percentage of carbon dioxide is different in inhaled and exhaled air. **(2)**

c Explain why exhaling is a form of excretion. **(2)**

d The following features can be seen in the lungs:

i thin membranes between the alveoli and the blood supply

7

ii a good blood supply

iii a large surface area.

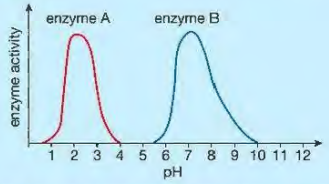
In each case explain how the feature helps gas exchange to happen quickly. **(6)**

Total 12 marks

SKILLS ANALYSIS 7

2 Digestion is brought about by enzymes converting large insoluble molecules into smaller soluble molecules that can be more easily absorbed.

a The activity of enzymes is influenced by pH and temperature. The graph shows the activity of two human enzymes from different regions of the gut at different pH values.



i Suggest which regions of the gut the two enzymes come from. Explain your answer. **(4)**

6

ii Which nutrient does enzyme A digest? **(1)**

b Farmers sometimes include urea in cattle food. The microorganisms in the rumen can use urea to make protein.

SKILLS CRITICAL THINKING 4

i In mammals, where in the body is urea made? **(1)**

5

ii What is urea made from? **(1)**

8

iii Suggest how feeding urea to cattle can result in an increased growth rate. **(1)**

SKILLS CRITICAL THINKING 7

BIOLOGY ONLY

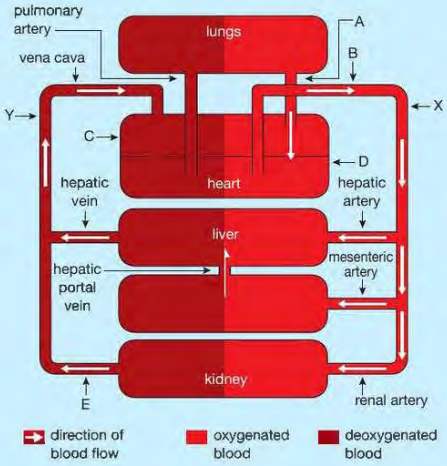
iv The Bowman's capsule and the loop of Henlé are both parts of a nephron. Explain how each of them helps to remove urea from the bloodstream. **(4)**

END OF BIOLOGY ONLY

Total 12 marks

SKILLS ANALYSIS 6

3 The diagram shows the human circulatory system. The system carries nutrients, oxygen and carbon dioxide around the body.



a Write down the names of the structures labelled A to E. **(5)**

7

b Give two differences between the blood vessels at point X and point Y. **(2)**

c During exercise, the adrenal gland releases the hormone adrenaline. Reflexes involving the medulla of the brain influence the heartbeat and breathing.

SKILLS CRITICAL THINKING 6

i Describe two effects of adrenaline on the heartbeat. **(2)**

ii Describe two other effects of adrenaline on the body. **(2)**

SKILLS REASONING 7

d How is a reflex action different from a voluntary action? **(2)**

6

e Following exercise there is a recovery period in which breathing rate and heart rate gradually return to pre-exercise levels. Explain why they do not return immediately to these levels. **(3)**

Total 16 marks

SKILLS INTERPRETATION 6 **4** The immune system responds to infections using white blood cells. A phagocyte is one type of white blood cell.

a Draw and label a phagocyte. **(3)**

SKILLS CRITICAL THINKING 5 **b** State one way that the structure of a phagocyte differs from that of a red blood cell. **(1)**

c Phagocytes carry out phagocytosis. Describe this process. **(2)**

6 **d** Describe how other white blood cells are involved in the immune response. **(3)**

Total 9 marks

SKILLS REASONING 7 **5** Humans and other mammals are able to maintain a constant body temperature which is usually higher than that of their surroundings.

a Explain the advantage in maintaining a constant, high body temperature. **(1)**

b The temperature of the blood is constantly monitored by the brain. If it detects a drop in blood temperature, the following things happen: the arterioles leading to the skin capillaries constrict, less sweat is formed and shivering begins.

8 **i** Explain how each response helps the body to keep warm. **(3)**

7 **ii** Explain how the structure of arterioles allows them to constrict. **(2)**

BIOLOGY ONLY

c When the weather is hot we produce less urine.

6 **i** What is the name of the hormone that controls the amount of urine produced by the body? **(1)**

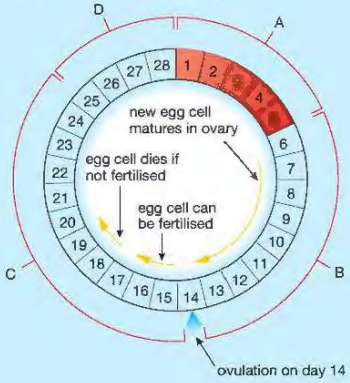
8 **ii** Explain why the body produces less urine in hot weather. **(1)**

iii Explain how the hormone in i) works in the kidney to produce less urine. **(3)**

END OF BIOLOGY ONLY

Total 11 marks

SKILLS ANALYSIS **6** The diagram represents a typical menstrual cycle.



a Using evidence from the diagram, answer the following questions.

During which of the stages A, B, C or D does:

SKILLS ANALYSIS 5 **i** the level of the hormone oestrogen increase in the blood

4 **ii** the level of the hormone progesterone increase in the blood

5 **iii** the uterine lining become more vascular

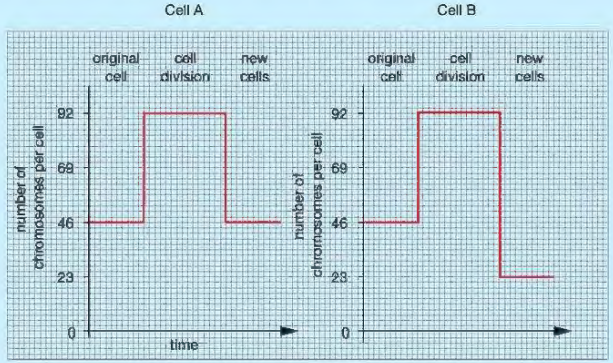
4 **iv** the levels of oestrogen and progesterone in the blood fall

v the uterine lining begin to break down? **(5)**

SKILLS REASONING 5 **b** Explain how knowledge of the menstrual cycle can be used to avoid pregnancy. **(3)**

Total 8 marks

SKILLS CRITICAL THINKING 8 **7** Cells can divide by mitosis or by meiosis. Human cells contain 46 chromosomes. The graphs show the changes in the number of chromosomes per cell as two different human cells undergo cell division.



a Which of the two cells, A or B, is dividing by meiosis? Explain how you arrived at your answer. **(3)**

b Explain the importance of meiosis, mitosis and fertilisation in maintaining the human chromosome number constant at 46 chromosomes per cell, generation after generation. **(6)**

7 **c** Give *three* differences between mitosis and meiosis. **(3)**

Total 12 marks

SKILLS EXECUTIVE FUNCTION 6 **8** Protein supplements are foods that some body-builders use to increase the growth of their muscles. Describe an investigation to find out if adding a protein supplement to the diet of rats will increase their growth.

Your answer should include experimental details and be written in full sentences. **(6)**